

RELATIONAL OPERATOR

```
a = 13
```

```
b = 33
```

```
# a > b is False
```

```
print(a > b)
```

```
# a < b is True
```

```
print(a < b)
```

```
# a == b is False
```

```
print(a == b)
```

```
# a != b is True
```

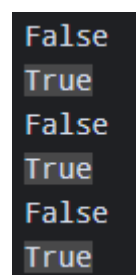
```
print(a != b)
```

```
# a >= b is False
```

```
print(a >= b)
```

```
# a <= b is True
```

```
print(a <= b)
```



```
False  
True  
False  
True  
False  
True  
False
```

```
x = 5
```

```
y = 3
```

```
print(x == y)
```

returns False because 5 is not equal to 3

```
x = 5
```

```
y = 3
```

```
print(x != y)
```

returns True because 5 is not equal to 3

```
x = 2
```

```
y = 3
```

```
print(x > y)
```

```
x = 3
```

```
y = 3
```

```
print(x >= y)
```

```
x = 5
```

```
y = 3
```

```
print(x <= y)
```

```
False
True
False
True
False
```

ASSIGNMENT OPERATORS

```
x = 5
x -= 3
print(x)
```

```
x = 5
x *= 3
print(x)
```

```
x = 5
x /= 3
print(x)
```

```
x = 5
x%=3
print(x)
```

```
x = 3
x **= 3
print(x)
```

```
2
15
1.6666666666666667
2
27
```

LOGICAL OPERATOR

```
x = 5
```

```
print(x > 3 and x < 10)
```

returns True because 5 is greater than 3 AND 5 is less than 10

```
x = 5
```

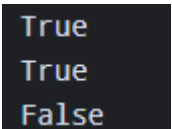
```
print(x > 3 or x < 4)
```

returns True because one of the conditions are true (5 is greater than 3, but 5 is not less than 4)

```
x = 5
```

```
print(not(x > 3 and x < 10))
```

returns False because not is used to reverse the result



```
True
True
False
```

MEMBERSHIP OPERATOR

```
x = ["apple", "banana"]
```

```
print("banana" in x)
```

returns True because a sequence with the value "banana" is in the list

```
x = ["apple", "banana"]
```

```
print("pineapple" not in x)
```

returns True because a sequence with the value "pineapple" is not in the list

```
True
True
```

IDENTITY OPERATOR

```
x = ["apple", "banana"]
```

```
y = ["apple", "banana"]
```

```
z = x
```

```
print(x is z)
```

returns True because z is the same object as x

```
print(x is y)
```

returns False because x is not the same object as y, even if they have the same content

```
x = ["apple", "banana"]
```

```
z = x
```

```
print(x is not z)
```

returns False because z is the same object as x

```
True
False
False
```